

DINOv3

Computer Vision und Mustererkennung



[SUBMITTED ON 13 AUG 2025]

AUTOREN

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Was macht Dinov3 besonders?

Pipeline

- Datenvorbereitung
- Large-Scale Training mit Self-Supervision
- **Gram Anchoring**
- Post-Training

Demo

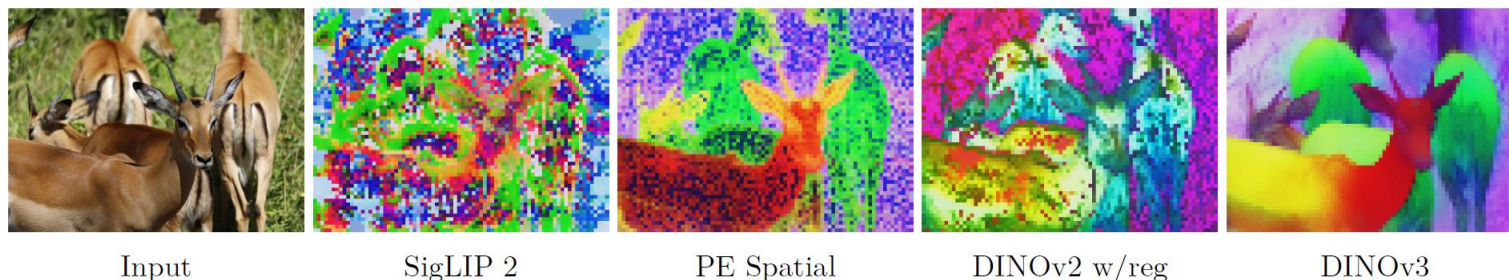
WAS MACHT DINOv3 BESONDERS?

dichte visuelle Repräsentationen

- ohne Fine-Tuning und ohne Text- oder Maskenannotation.
- Reines Self-Supervised Learning (keine Labels, kein Text nötig)

Neu:

- **Gram Anchoring** verhindert Feature-Degradation



Input

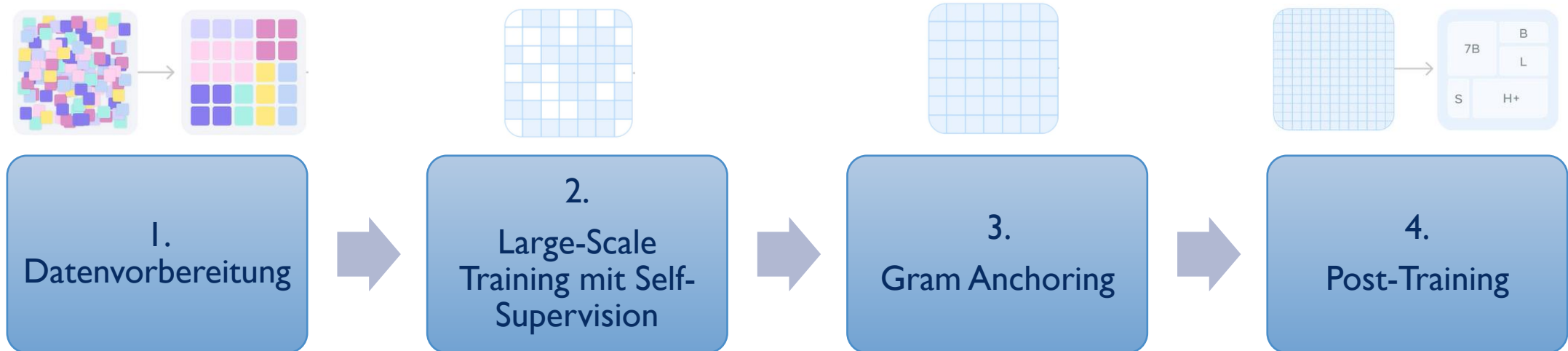
SigLIP 2

PE Spatial

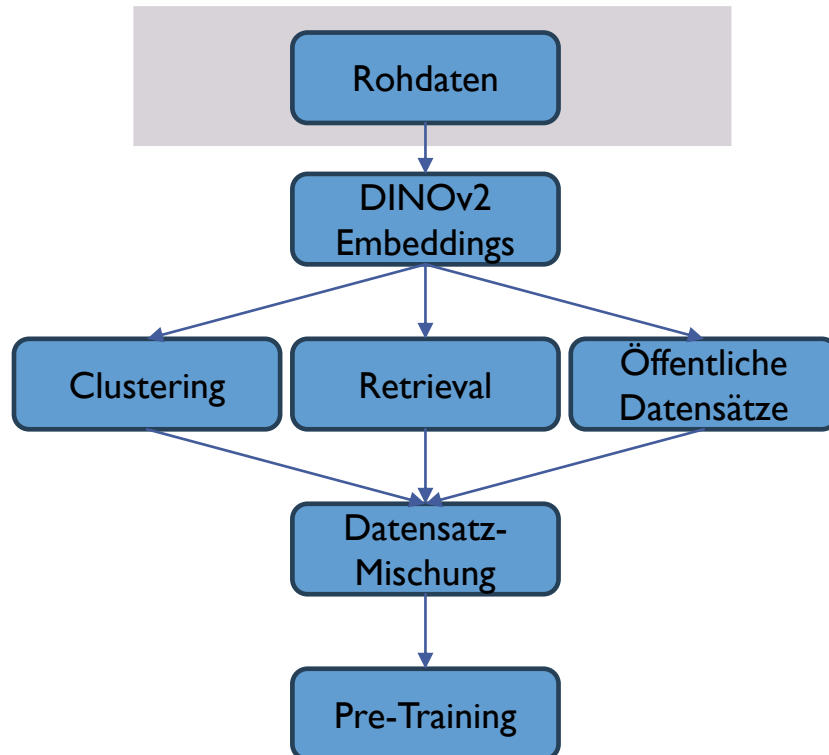
DINOv2 w/reg

DINOv3

PIPELINE



DATENVORBEREITUNG

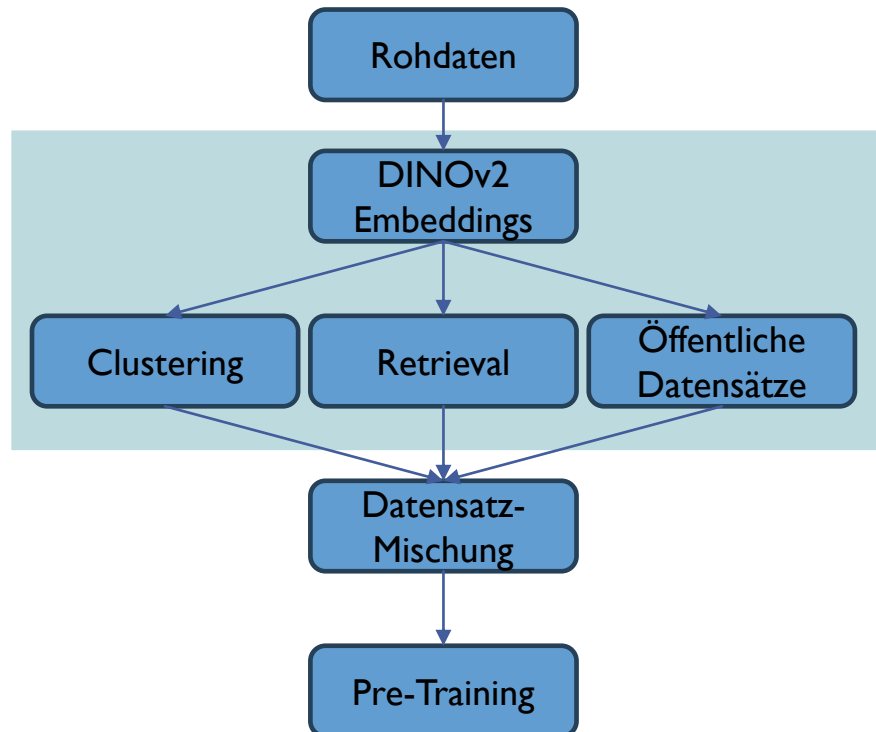


Datenkollektion

- großen Datenpool
 - Instagram

→ ca. 17 Milliarden Bildern.

DATENVORBEREITUNG

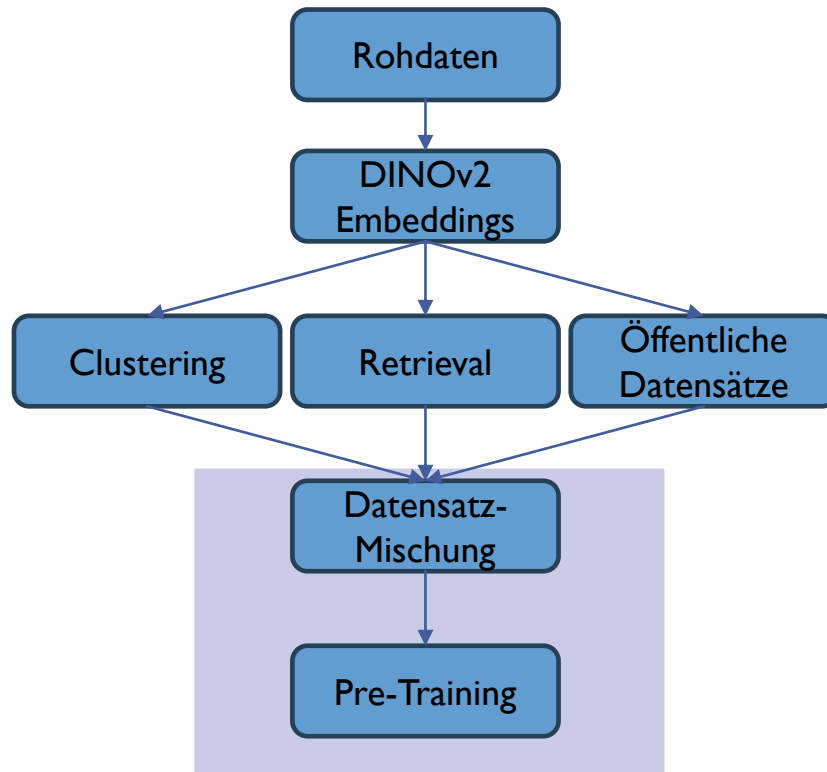


Datenkuratierung

3 Datenset-Teile:

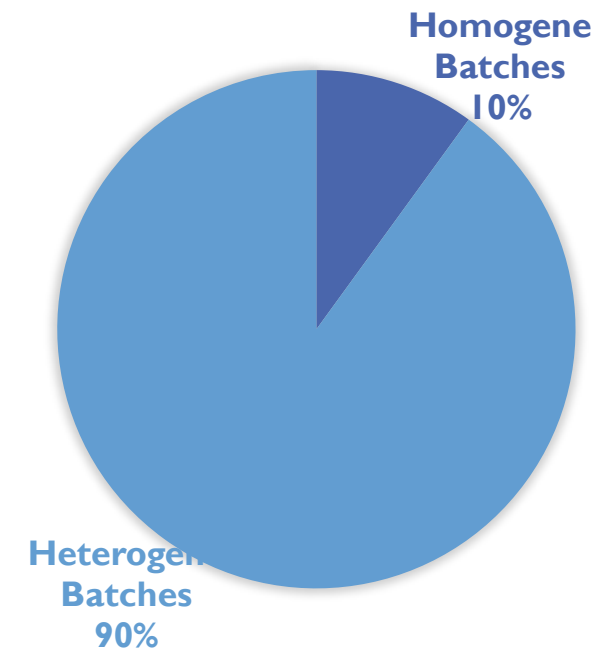
- Clustering-basierte Kuratierung (basierend auf k-Means)
- Retrieval-basierte Kuratierung
- Verwendung von öffentlich zugänglichen Rohdaten (ImageNet1k, ImageNet22k, Mapillary Street-level Sequences)

DATENVORBEREITUNG

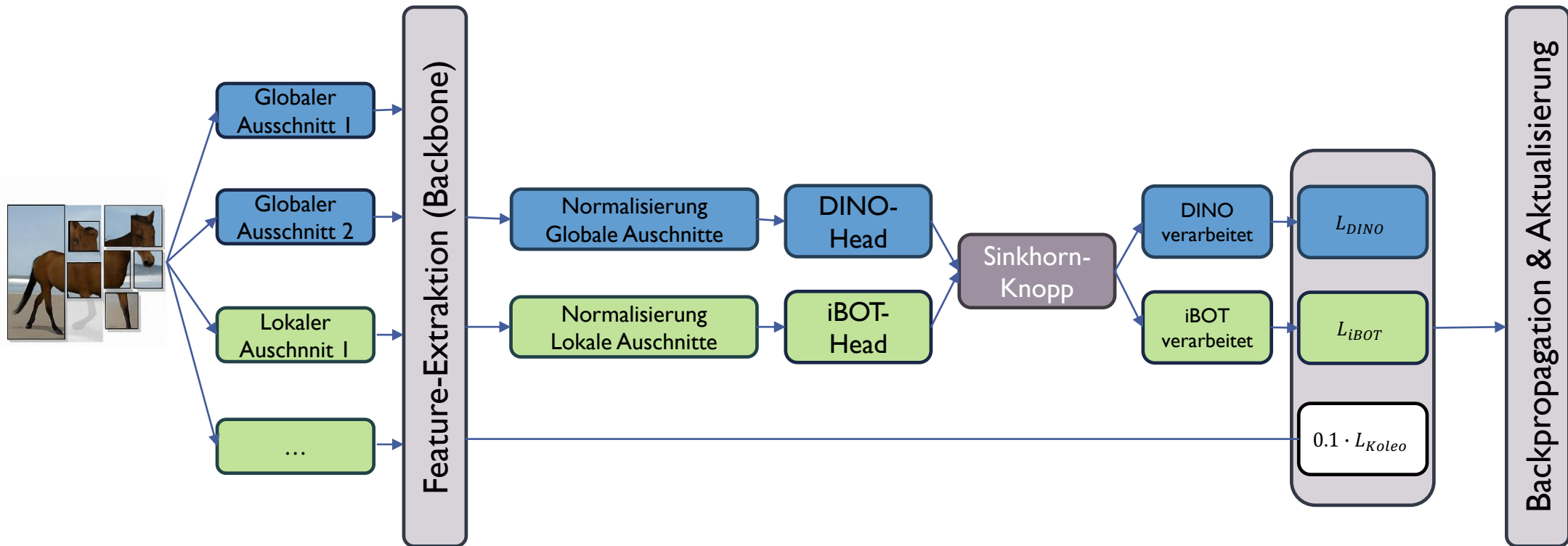


Daten-Sampling

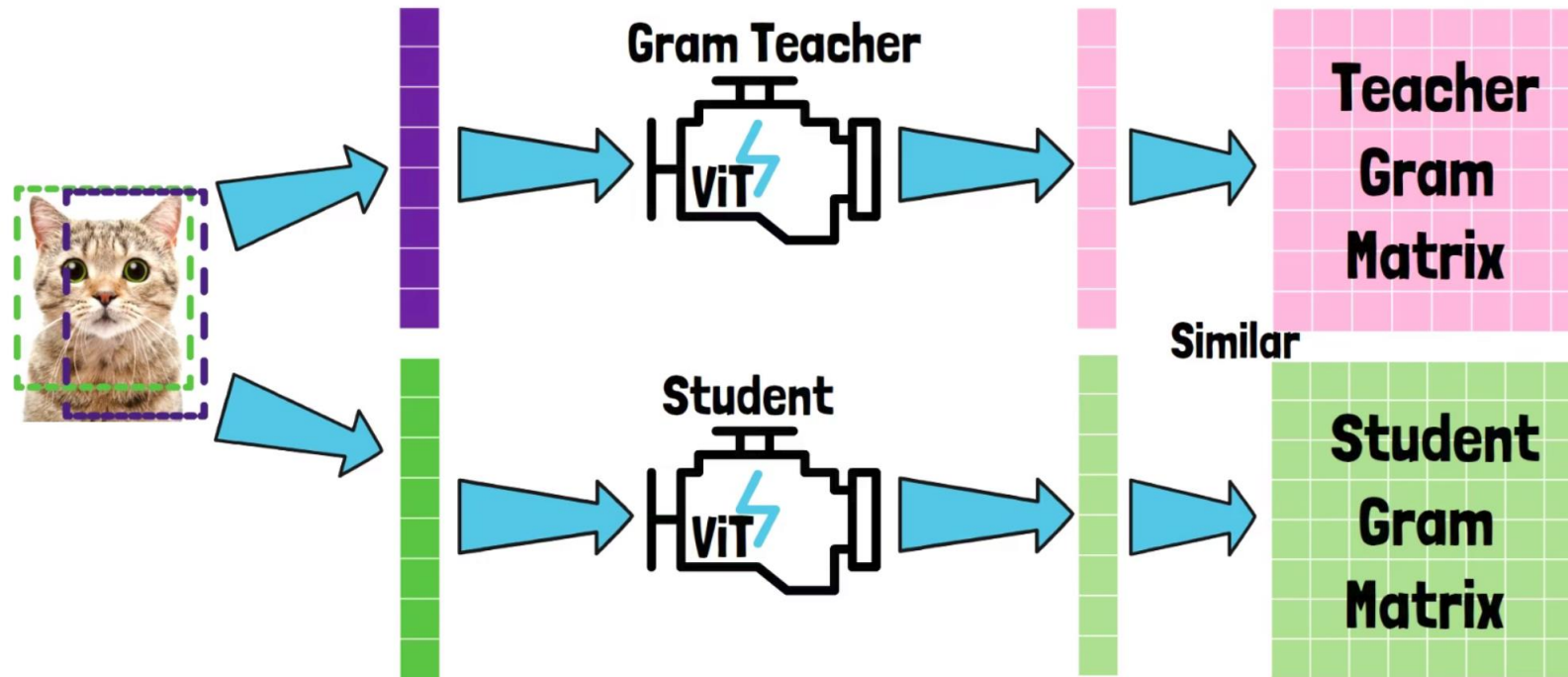
- Homogene Batches (nur ImageNet1k)
- Heterogene Batches (Mix aus alle)



LARGE-SCALE TRAINING MIT SELF-SUPERVISION

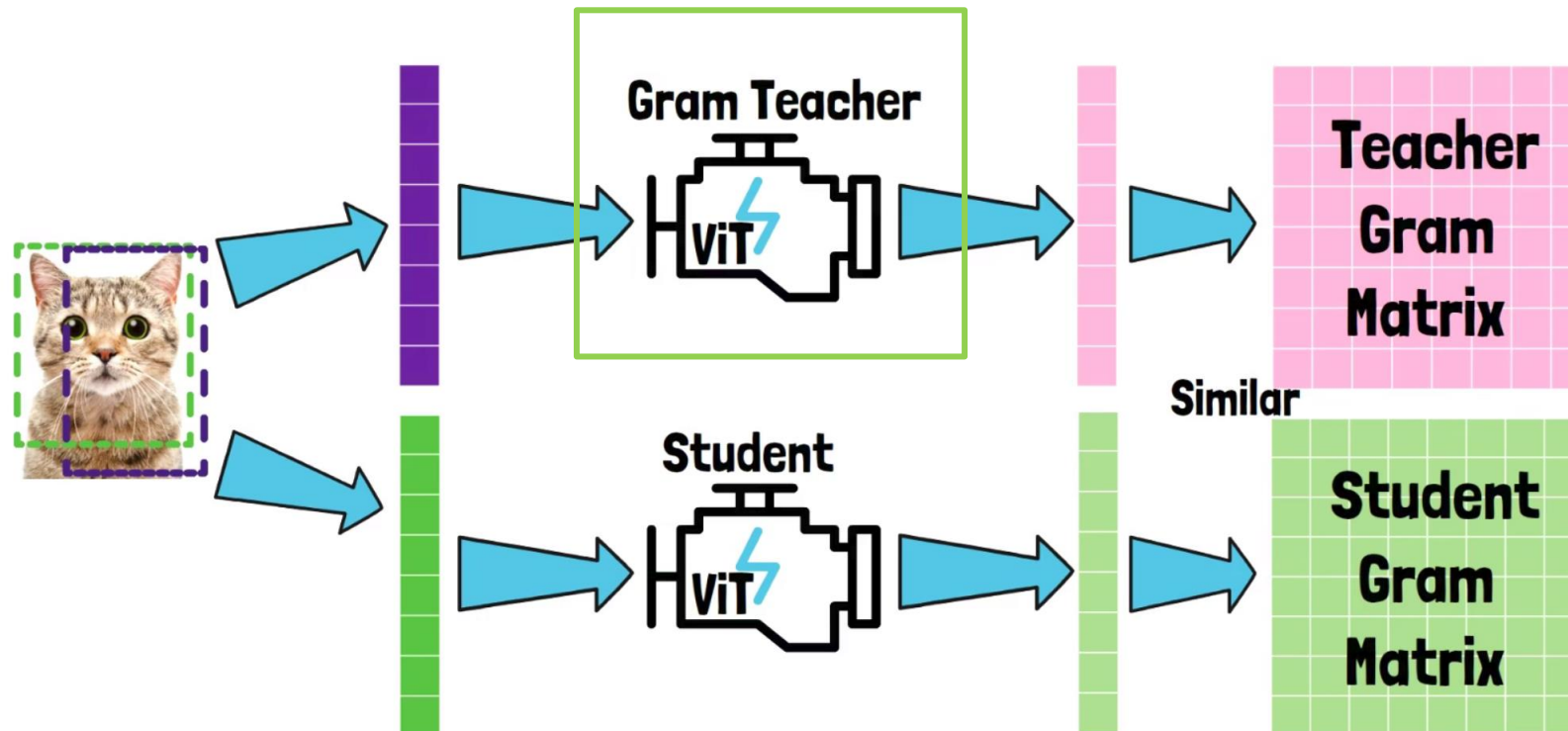


GRAM ANCHORING

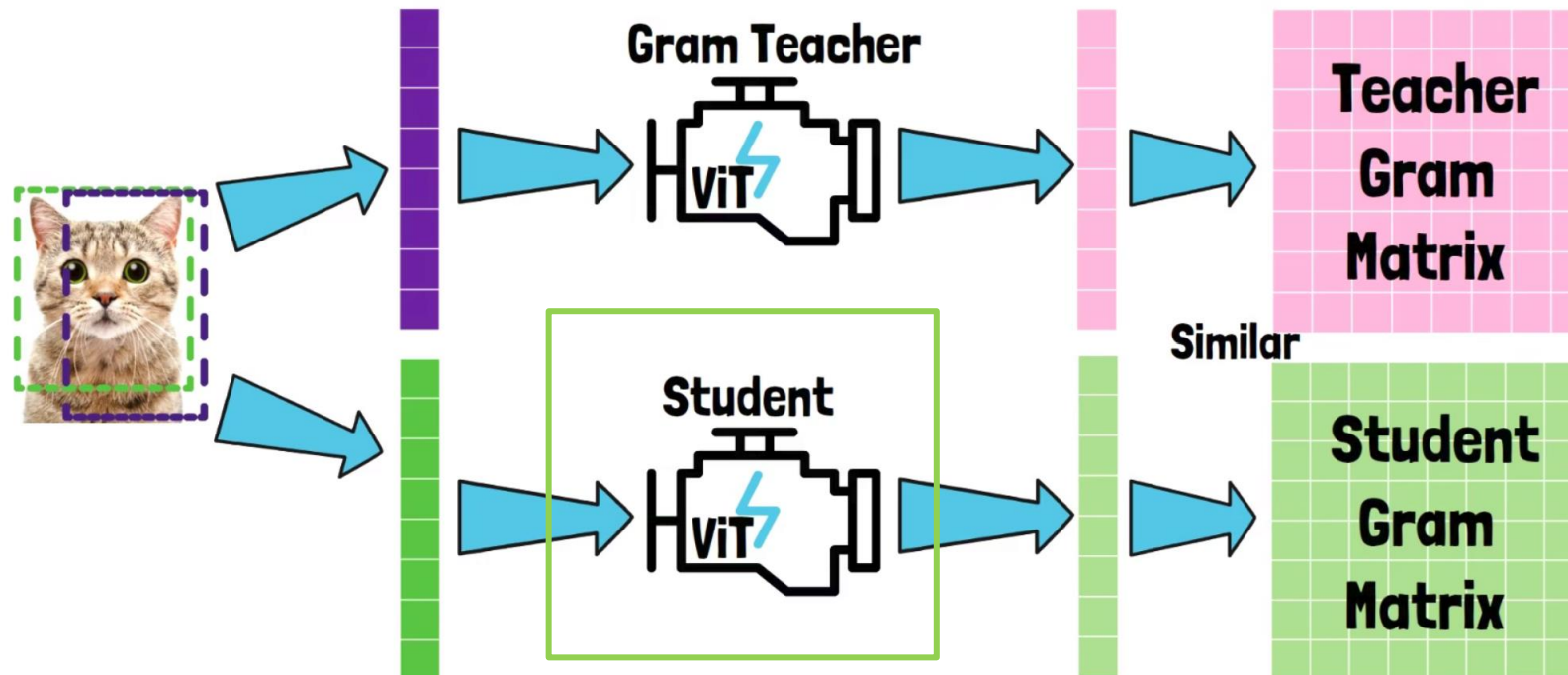


Ziel → verhindert Verschlechterung der Konsistenz auf Patch-Ebene

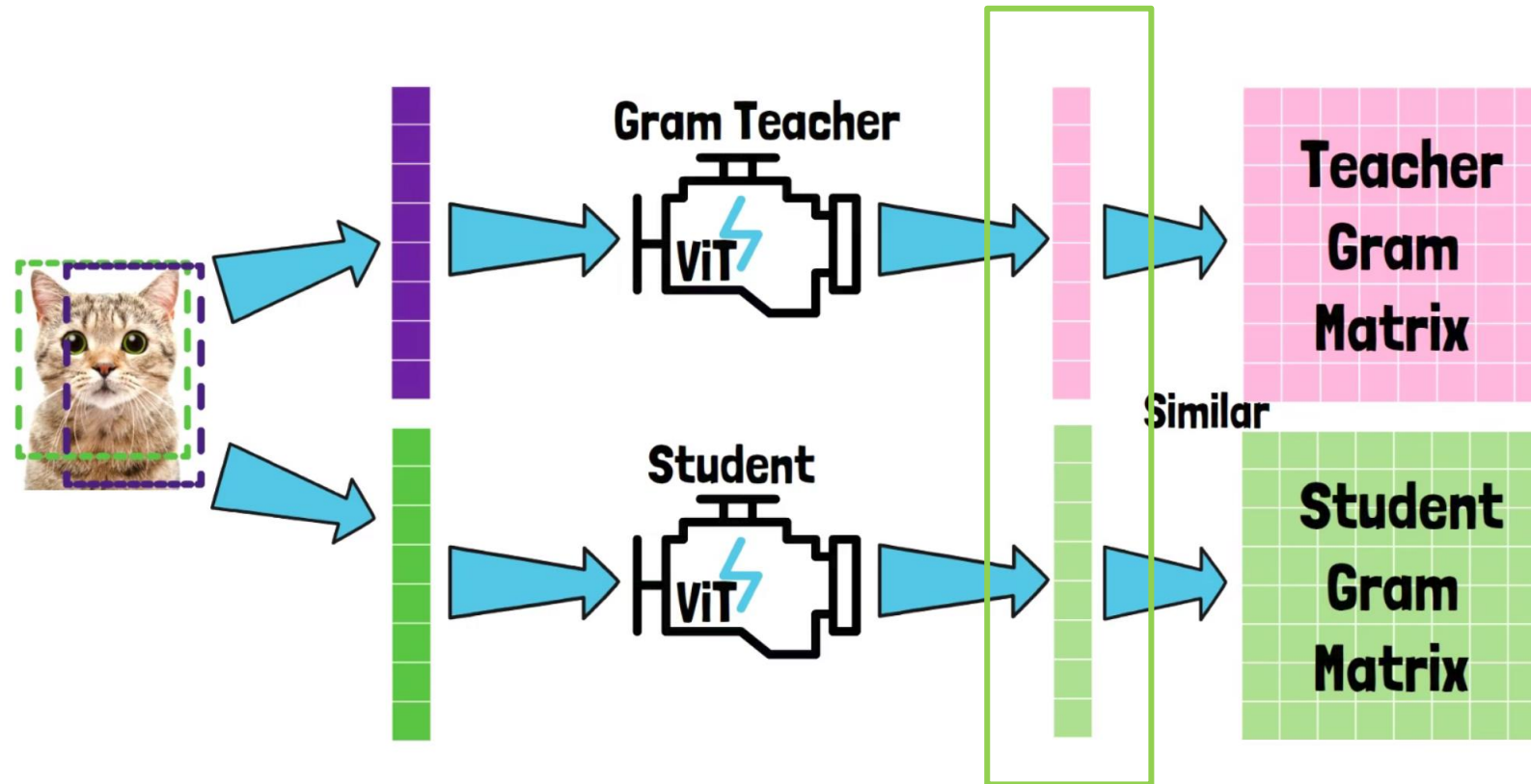
GRAM ANCHORING



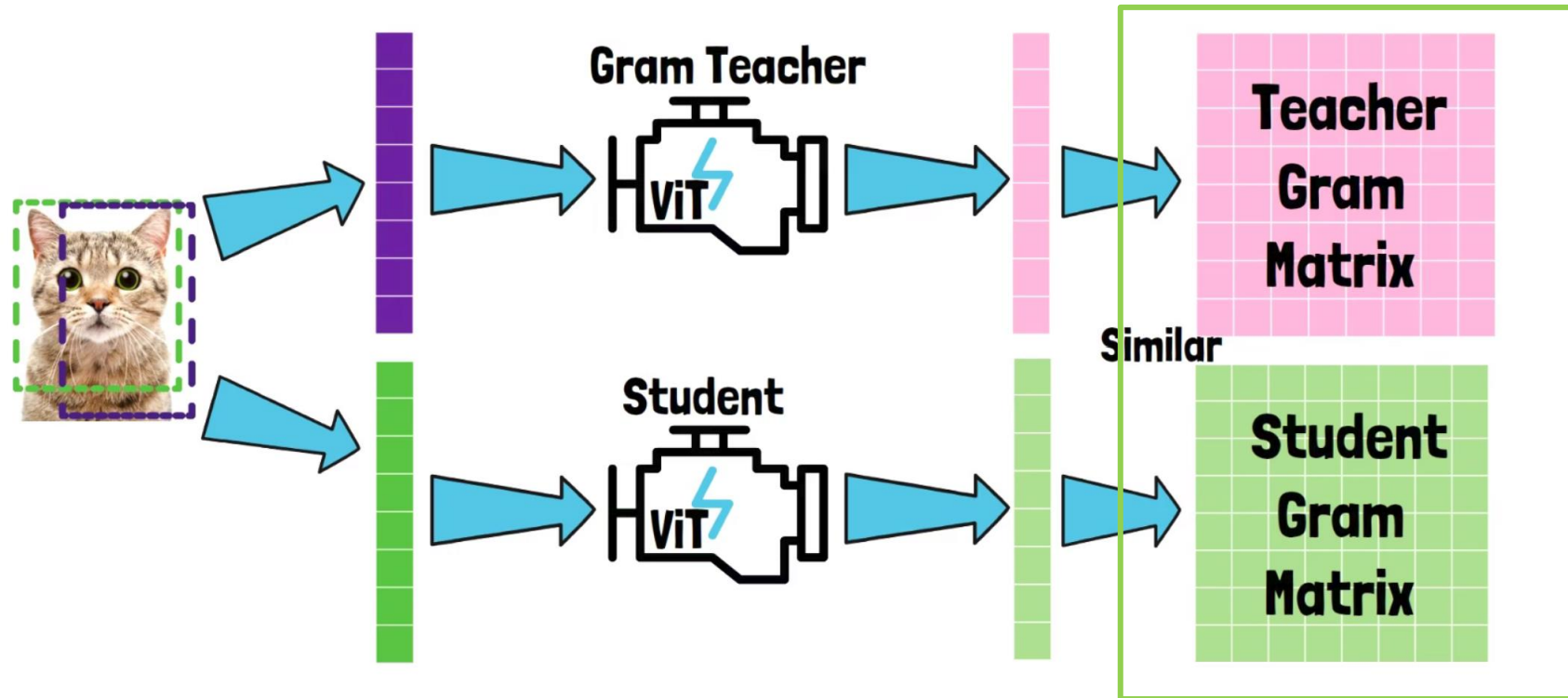
GRAM ANCHORING



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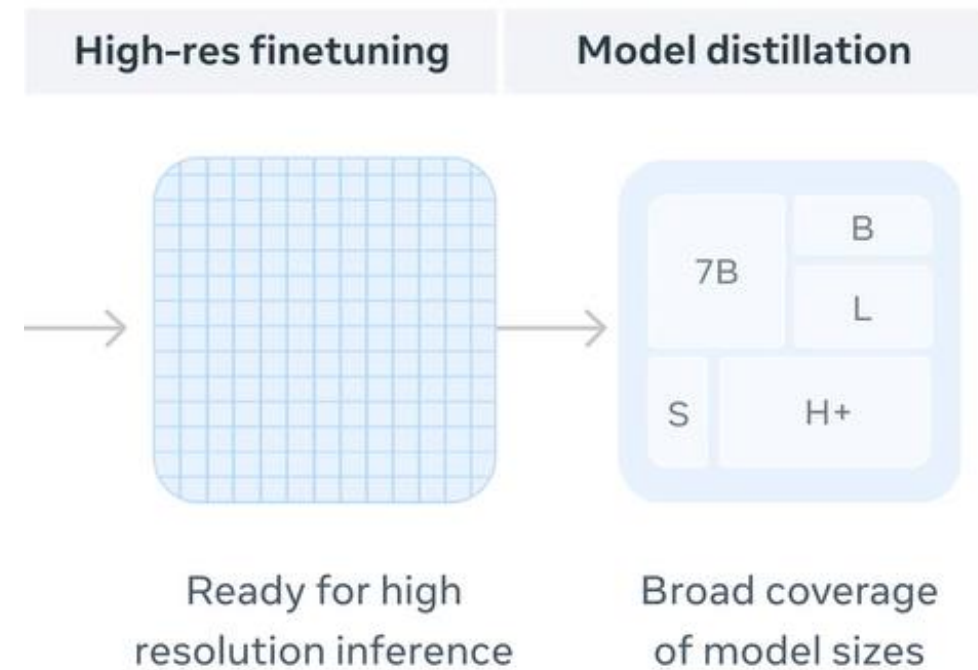
GRAM ANCHORING



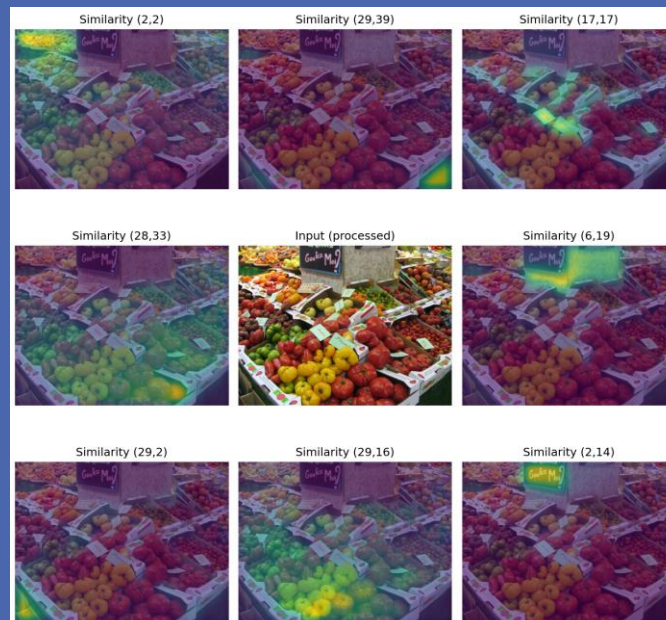
POST-TRAINING



- Hochauflösende Anpassung (Auflösungs-Skalierung)
- Model Destillation
- DINOv3 mit Text abgleichen (optional)



DEMO



VIELEN DANK FÜR EURE AUFMERKSAMKEIT

